Benchmark report: synthetic GARCH daily series (SYNTH-GARCH)

QuantEval v0.1.0 · run `51e2fd563c651f52` · generated 2026-10-10T15:33:34 UTC

Research evaluation only. This report measures statistical properties of forecasting models on historical OHLCV data supplied for the evaluation. It is not investment advice, makes no statement about any security's future price, and contains no instructions to trade. Simulated historical results (section 6) depend on stated cost and execution assumptions and do not indicate future results.

1. Executive summary

- Across 134 non-overlapping out-of-sample origins and a 5-bar horizon, the lowest MAE (price units, all steps) came from **arima** (1.1754).
- **drift**: MAE 1.017x the naive random walk (Diebold-Mariano p=0.147 at bar 5); directional accuracy 48.5% (95% CI 40.2%-56.9%, n=134), not statistically different from 50%.
- **arima**: MAE 0.998x the naive random walk (Diebold-Mariano p=0.586 at bar 5); directional accuracy 54.5% (95% CI 46.0%-62.7%, n=134), not statistically different from 50%.
- **kronos_mini**: MAE 1.196x the naive random walk (Diebold-Mariano p=0.002 at bar 5); directional accuracy 49.3% (95% CI 40.9%-57.6%, n=134), not statistically different from 50%.
- Forecast-accuracy results (this section) are reported separately from the research backtest (section 6); a model can be more accurate without producing a better simulated outcome, and vice versa.

2. Data card

File	synthetic_daily.csv	**SHA-256**	`37750b5c7366ec119e73902469959022916cc93c58fbf018dfcf6a74db56b6a0`
Instrument	SYNTH-GARCH	**Rows**	800
Span (UTC)	2024-01-01 → 2026-03-10	**Bar spacing**	86400s (median 24.0 h), includes weekend bars
Columns	timestamp, open, high, low, close, volume	**Provenance**	Fully synthetic; generated by examples/synthetic_ohlcv.py (rows=800, seed=7). No real market data.

3. Protocol

- Expanding-window walk-forward: 134 origins from 2024-05-09 to 2026-03-05, step 5 bars (evaluation windows do not overlap), embargo 0 bar(s) on parameter estimation, minimum history 128 bars.
- At each origin every model receives only bars up to and including the origin; forecasts cover the next 5 bars of close. Future timestamps given to Kronos are generated from the bar calendar, not read from the file.
- Models:
  - **naive**: Random walk: every future close equals the last observed close.
  - **drift**: Random walk with drift: mean log-return over the lookback window, extrapolated.
  - **arima**: ARIMA(1, 1, 1) on log close, refit every 1 origin(s)
  - **kronos_mini**: Kronos-mini zero-shot (4,108,032 params), context 256, mean of 5 samples, T=1.0, top_p=0.9, CPU — weights `NeoQuasar/Kronos-mini@f4e6869`, tokenizer `NeoQuasar/Kronos-Tokenizer-2k@26966d0`
- Metrics: MAE / RMSE / MAPE over all 5 steps; relative MAE vs naive; Diebold-Mariano test on squared errors at bar 5 vs naive; directional accuracy of the bar-5 move with Wilson 95% CI and a two-sided binomial test vs 50% (origins where the model or the market shows no change are excluded).

4. Forecast accuracy (out-of-sample)

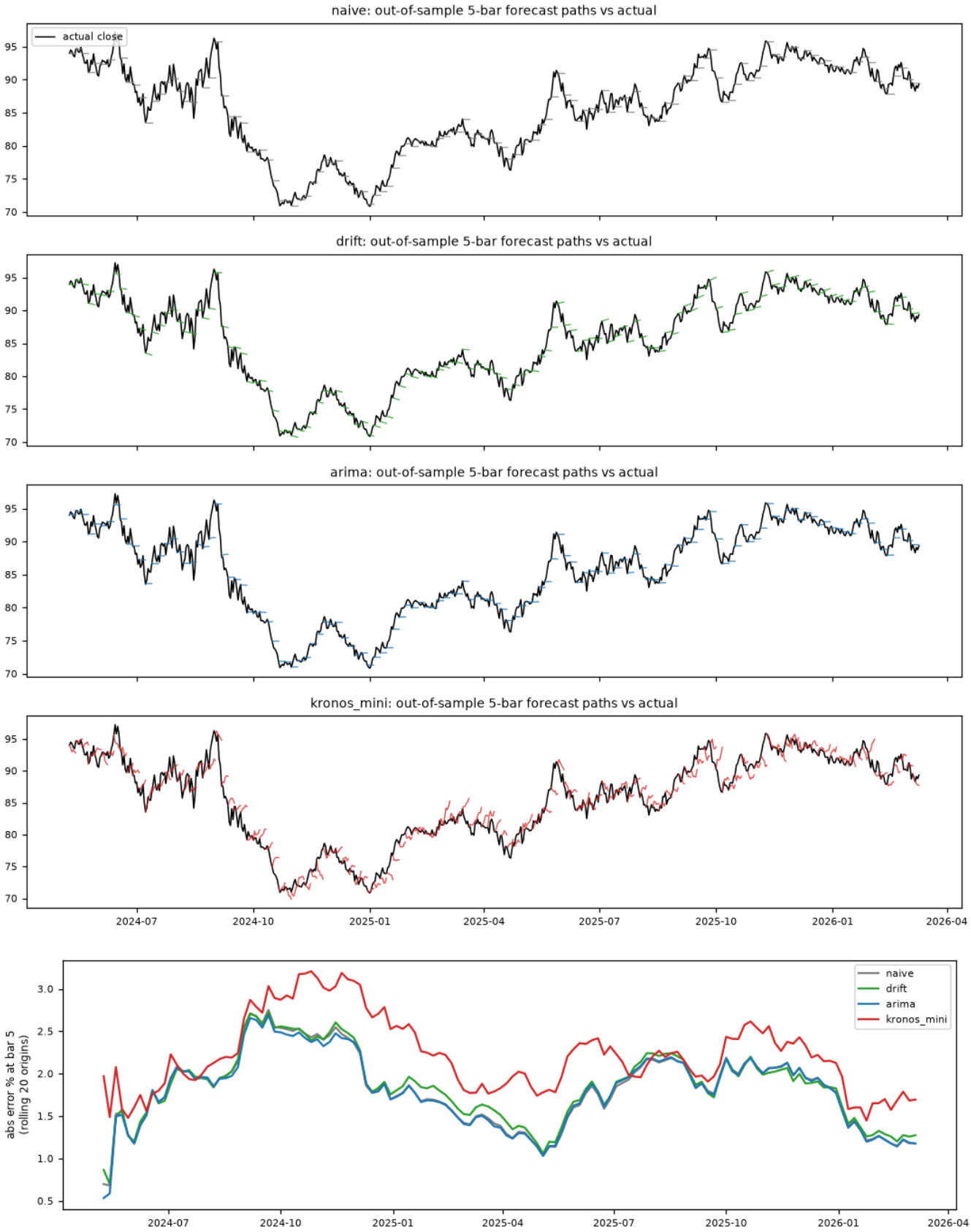
Model	MAE	RMSE	MAPE	MAE / naive	DM stat	DM p	Dir. acc.	95% CI	n	p vs 50%
naive	1.1773	1.5558	1.371%	1.0000	n/a	n/a	n/a	n/a	0	n/a
drift	1.1977	1.5755	1.395%	1.0173	1.457	0.147	48.5%	40.2%–56.9%	134	0.796
arima	1.1754	1.5516	1.369%	0.9984	-0.546	0.586	54.5%	46.0%–62.7%	134	0.342
kronos_mini	1.4083	1.8198	1.655%	1.1962	3.172	0.002	49.3%	40.9%–57.6%	134	0.931

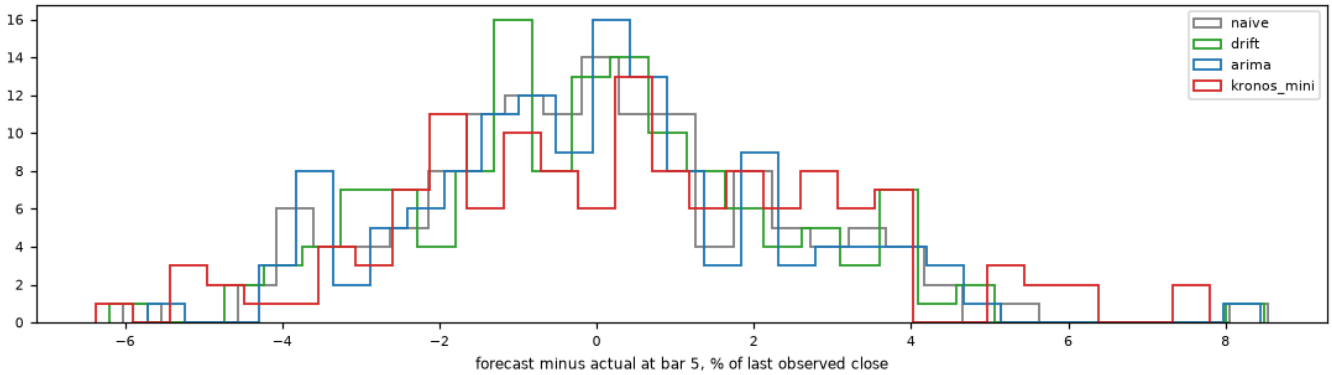
Inference latency (measured on this run's CPU)

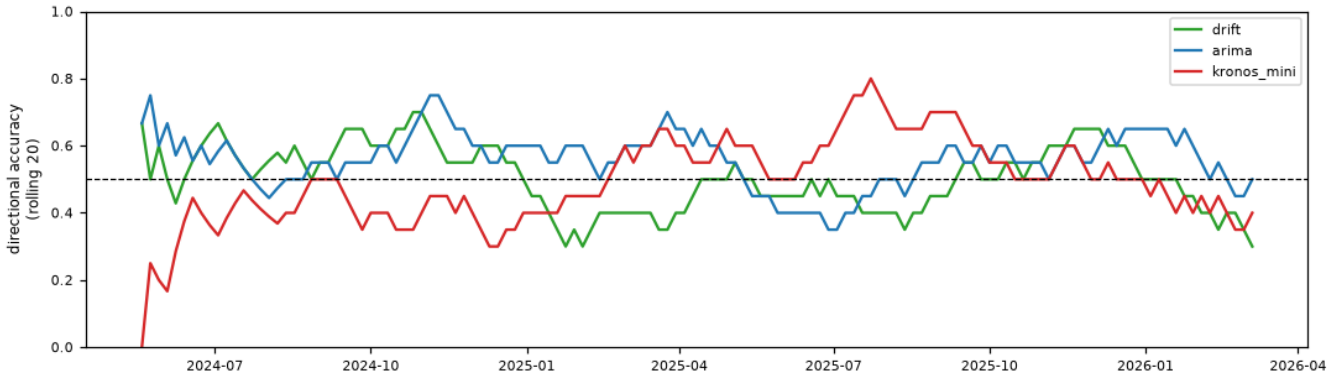
Model	forecasts	p50 ms	p95 ms	mean ms	total s
naive	134	0.04	0.05	0.04	0.01
drift	134	0.07	0.09	0.07	0.01
arima	134	31.55	79.69	37.00	4.96
kronos_mini	134	166.07	277.91	188.08	25.20

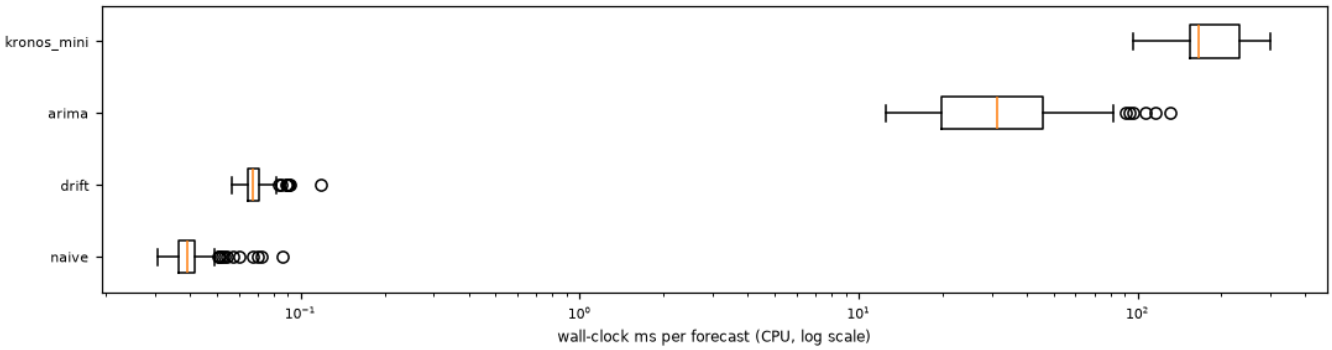
Hardware: Intel(R) Xeon(R) Processor, 8 logical CPUs, torch threads 4.

5. Charts (out-of-sample)









6. Research backtest (historical simulation, not investment advice)

Rule evaluated for each model: at each origin, take exposure equal to the sign of the model's forecast move to bar 5 (long_short), entered 1 bar(s) after the origin close and held for 5 bars; costs 5.0 bps per side (sensitivity at 0, 5, 10, 25 bps). No slippage model beyond the bps assumption, no funding/borrow costs, no position sizing. The section is shown only when at least 30 active non-overlapping periods exist and all leakage checks pass.

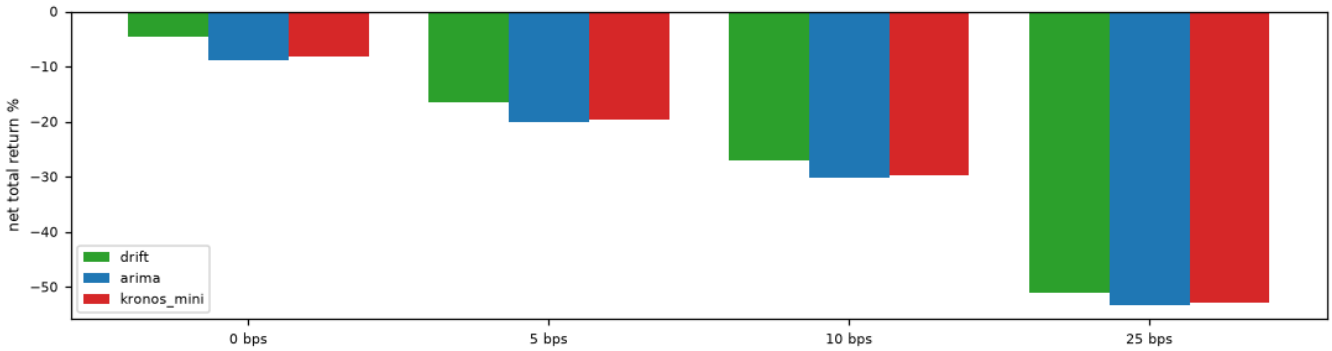
Model	Status	active / periods	gross return	net return	ann. Sharpe (net)	max DD (net)	hit rate (net)
naive	suppressed: only 0 non-overlapping active periods (< 30 required)						
drift	shown	133 / 133	-4.7%	-16.6%	-0.43	-33.3%	45.1%
arima	shown	133 / 133	-8.8%	-20.2%	-0.56	-31.2%	49.6%
kronos_mini	shown	133 / 133	-8.3%	-19.7%	-0.54	-36.0%	47.4%

Passive long reference over the same span (one round trip of costs): gross -5.8%, net -5.9%.

Cost sensitivity (net total return)

Model	0 bps	5 bps	10 bps	25 bps
drift	-4.7%	-16.6%	-27.0%	-51.1%
arima	-8.8%	-20.2%	-30.1%	-53.2%
kronos_mini	-8.3%	-19.7%	-29.7%	-52.9%





7. Leakage and data-quality checks

Check	Result
Fold structure (fit end < target start, no overlapping windows)	passed
Negative control: deliberately leaky model must be detected	detected (detector working); max forecast change 6.4938
Future-invariance: naive (origins 129, 464, 794)	passed (max forecast change when future bars are altered: 0.0000)
Future-invariance: drift (origins 129, 464, 794)	passed (max forecast change when future bars are altered: 0.0000)
Future-invariance: arima (origins 129, 464, 794)	passed (max forecast change when future bars are altered: 0.0000)
Future-invariance: kronos_mini (origins 129, 464, 794)	passed (max forecast change when future bars are altered: 0.0000)
Pre-training window (Kronos)	not applicable: synthetic data

Warnings

No data-quality warnings.

8. Limitations

- One instrument, one horizon, one historical period. Results may not transfer to other instruments, regimes or horizons.
- Kronos is evaluated zero-shot with fixed sampling settings; other settings, larger variants or fine-tuning may behave differently. Its training data is not public, so contamination cannot be ruled out for periods before its release.
- ARIMA uses a fixed order on log prices; no order search was performed.
- Statistical tests are not adjusted for multiple comparisons across models.
- The research backtest ignores slippage beyond the bps assumption, funding, borrow, market impact and capacity.

9. Reproducibility

Run id	`51e2fd563c651f52`	**Code commit**	`d1295a5a161246f25ff2da027279741df4af3e68`
Seed	42 (Kronos: seed + origin index)	**predictions.csv SHA-256**	`0511ac634a9f5b61450ea7eadfc2aa31dfb07ef21adc10434a1897fc818f2bd8`
Packages	quanteval 0.1.0; numpy 2.3.5; pandas 2.3.3; scipy 1.18.1; statsmodels 0.15.0; torch 2.14.1+cpu; einops 0.8.1; huggingface_hub 0.33.1; safetensors 0.6.2; matplotlib 3.11.2; jinja2 3.1.6; weasyprint 70.0; pyarrow 26.0.0; pyyaml 6.0.3;		
Timings (s)	load_validate_s=0.01; negative_control_s=0.0; model_load_s=1.33; leakage_checks_s=1.38; walk_forward_s=30.23; total_before_report_s=32.99;		

The full configuration, hashes, timings and the compute-cost estimate are in `manifest.json`. Re-run with `quanteval reproduce --manifest manifest.json --csv <same file>`.

Third-party notices: Kronos model code © 2025 ShiYu, MIT License (vendored at commit 67b630e); Kronos-mini and Kronos-Tokenizer-2k weights published under MIT on Hugging Face by NeoQuasar. Citation: Shi et al., "Kronos: A Foundation Model for the Language of Financial Markets", arXiv:2508.02739.